

TIKHORETSK, Russian Federation

WMO: 348380

Lat: 45.850N

Lon: 40.083E

Elev: 255

StdP: 14.56

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.3	9.8	-2.2	4.9	5.1	3.8	6.7	10.8	18.7	34.5	16.2	31.9	5.9	10	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	95.2	70.9	91.9	69.5	88.6	68.4	74.3	87.6	72.4	85.6	70.8	83.9	6.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	112.1	80.4	68.2	104.5	78.5	66.5	98.5	76.8	38.0	88.0	36.2	85.3	34.9	83.9	89.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.1	13.9	12.1	DB	-1.5	99.3	10.1	2.8	-8.7	101.3	-14.6	102.9	-20.3	104.5	-27.6	106.5	(4)
(5)			WB	-2.7	78.7	9.5	4.8	-9.6	82.1	-15.2	84.9	-20.6	87.6	-27.5	91.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.2	30.1	32.2	41.2	53.1	63.6	70.8	76.5	75.6	65.6	53.2	41.6	33.8		
	DBStd	18.32	10.54	10.35	7.82	7.35	7.27	5.94	5.48	6.29	6.72	8.87	8.79	9.53		
	HDD50	2307	619	499	290	49	2	0	0	0	2	67	274	505		
	HDD65	5358	1083	918	739	363	116	15	0	3	72	379	702	968		
	CDD50	3479	0	1	16	141	425	624	821	793	469	165	22	2		
	CDD65	1057	0	0	0	5	73	189	356	332	90	12	0	0		
	CDH74	10457	0	0	1	60	665	1661	3757	3519	699	95	0	0		
	CDH80	4242	0	0	0	7	167	579	1680	1605	183	21	0	0		
	WSAvg	5.6	6.3	6.5	7.0	6.4	5.2	4.7	4.6	4.7	5.0	5.2	5.8	5.9		
	PrecAvg	24.5	2.1	1.4	1.5	1.9	2.5	3.0	2.3	1.8	1.6	1.6	2.2	2.6		
(15) Precipitation	PrecMax	36.6	5.5	4.8	3.3	5.4	6.6	8.6	5.3	4.6	3.1	3.5	4.6	7.1		
	PrecMin	16.3	0.2	0.2	0.0	0.0	0.5	0.0	0.1	0.1	0.0	0.1	0.2	0.3		
	PrecStd	4.0	1.3	0.9	0.8	1.2	1.3	1.9	1.3	1.3	0.8	0.8	1.3	1.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.8	60.5	69.2	80.2	89.2	93.6	98.7	100.1	90.3	82.4	68.1	57.5		
		MCWB	46.2	50.2	52.6	59.0	66.8	70.3	72.9	70.3	70.5	63.2	55.2	49.8		
	2%	DB	48.2	52.7	62.3	74.3	84.5	89.9	95.4	95.6	84.9	75.6	61.2	51.8		
		MCWB	43.7	45.8	49.6	57.3	64.6	68.7	71.5	70.7	66.6	61.3	53.8	46.6		
	5%	DB	44.6	48.6	57.2	69.9	81.1	86.4	92.1	92.1	81.3	70.5	56.9	48.5		
		MCWB	41.1	43.3	47.6	55.2	63.4	67.4	69.9	69.3	65.5	59.4	51.6	44.6		
	10%	DB	41.7	45.0	52.8	66.1	77.7	82.9	88.6	88.4	77.8	65.8	53.1	45.5		
		MCWB	39.0	40.9	45.5	53.5	62.1	66.8	68.9	68.2	63.7	57.4	48.9	42.5		
	0.4%	WB	47.6	50.6	54.6	61.1	69.5	75.3	78.3	77.2	73.7	64.7	56.8	51.0		
		MCDB	51.0	58.3	64.9	74.0	83.9	85.1	91.8	88.3	87.1	77.5	63.5	55.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	44.1	46.4	51.2	58.6	67.3	72.0	74.7	73.7	69.2	62.6	54.5	47.3		
		MCDB	47.4	52.2	60.4	71.2	80.5	84.7	88.5	87.5	81.1	73.7	60.6	51.2		
	5%	WB	41.4	43.7	48.6	56.7	65.3	70.3	72.5	71.9	67.1	60.4	52.0	44.9		
		MCDB	44.4	48.4	55.7	67.0	77.4	82.5	85.7	85.6	77.4	68.4	56.5	48.2		
	10%	WB	39.0	41.0	46.0	54.9	63.4	68.4	70.9	70.3	65.3	58.3	49.2	42.6		
		MCDB	41.5	44.8	52.2	63.8	74.4	79.3	84.0	83.5	74.8	64.9	52.6	45.4		
	5% DB	MDBR	10.2	12.2	14.2	17.6	18.3	18.2	19.5	20.3	19.0	15.8	12.1	9.8		
		MCDBR	13.2	17.4	22.5	23.9	22.9	22.1	22.9	24.2	23.4	22.8	18.6	14.0		
		MCWBR	10.0	11.7	12.7	11.1	10.0	8.9	9.2	9.7	10.9	11.6	11.6	10.6		
		MCWBR	12.9	16.5	20.0	20.4	20.0	20.1	20.2	20.6	20.7	20.5	17.0	13.3		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.1	11.6	12.1	10.4	9.5	10.3	10.3	10.4	11.1	11.1	11.2	10.5		
	Clear-Sky Solar Irradiance	taub	0.346	0.378	0.418	0.456	0.420	0.428	0.456	0.452	0.420	0.389	0.355	0.348		
		taud	2.262	2.179	2.117	2.061	2.251	2.260	2.193	2.201	2.266	2.311	2.340	2.275		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	231	246	253	254	269	267	258	253	251	240	227	215		
		Edh at Noon	27	35	43	49	42	42	44	43	36	30	24	24		
(44) All-Sky Solar Radiation	RadStd		377	656	968	1490	1881	2044	2046	1841	1347	834	450	308		
			64	95	87	147	191	126	99	119	129	101	61	40		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (7 neighbors)		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+178	+108	

Nomenclature: See separate page